Question 1

$$\frac{9}{20}by + \frac{7}{15} = -\frac{9}{20}ax$$

$$\frac{1}{8}x + \frac{1}{18}y = \frac{a}{2} + \frac{5}{12}x$$

In the given system of equations, *a* and *b* are constants and $a > 0$. The system has infinitely many solutions. What is the value of $a - b$?

Question 2

$$P = \frac{V^2}{R}$$

The equation above gives the power in an electrical circuit, P, in Watts as a function of the voltage, V, in Volts and the resistance, R, in Ohms. An electrical engineer is measuring two circuits. The first circuit has a voltage of v Volts and a resistance of r Ohms. The second circuit has a voltage of $12v$ Volts and a resistance of $3r$ Ohms. The numbers v and r are positive constants. Which of the following could represent the ratio of the power in the second circuit to the power in the first circuit?

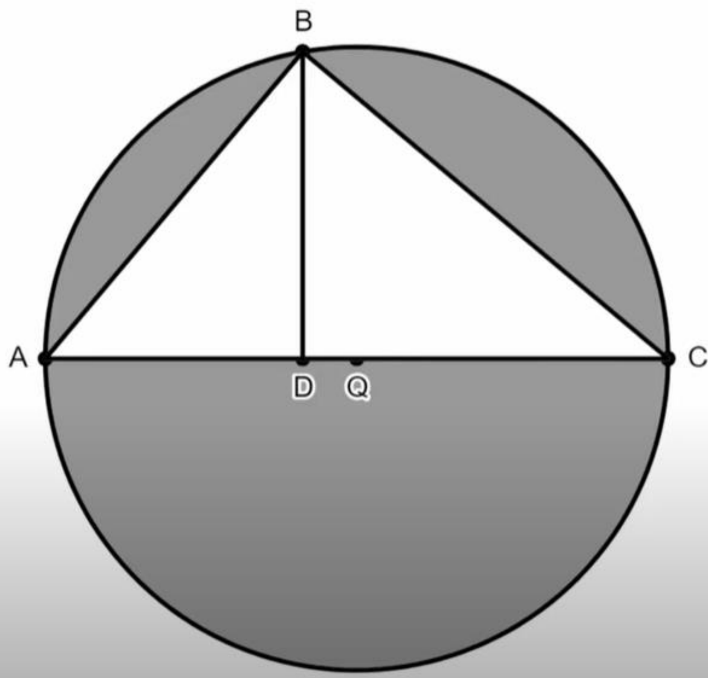
A) 3 to 1

B) 4 to 1

C) 48 to 1

D) 81 to 1

Question 3



Point Q is the center of the circle. Point A, point B, and point C all lie on the circle. Point D lies on segment AC as shown. Segment BD is perpendicular to segment AC. In the figure shown, $AB = 58$ and $AD = 40$. If a point within the figure is selected at random, what is the probability that the point is within the shaded region? (Express your answer as a decimal or fraction, not as a percent.) Round your answer to the nearest thousandth.

Question 4

x	y
-7	$d+3$
-4	$4k+6d$
$1-3d$	8

The table shows three values of x and their corresponding values of y for the linear relationship between x and y. The x-intercept of the line occurs when $x = -4$. The number d is a negative constant. The number k is a constant. What is the value of dk?

Question 5

$$f(x) = ax^2 - 7x + c$$

The function *f* is defined above, where *a* and *c* are constants. The graph of $y = f(x)$ is a parabola that has a vertex at the point (h, k), where *h* and *k* are constants. If $f(-9) = f(-2)$ and $c > 0$, which of the following must be true?

I. $a < -\frac{3}{4}$

II. $k > 16$

A) I only B) II only C) I and II D) Neither I nor II

Question 6

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. A triangle has side lengths of $4x + 20$, $2x + 8$, and $7x - 11$. Which inequality represents all possible values of x?

A) $x < 39$

B) $8 < 2x < 78$

C) $3 < x < 38$

D) $23 < 5x < 195$

Question 7

A circle is centered at Point O with coordinates $(0, 0)$ in the xy-plane. The circle contains point A with coordinates $(25, 0)$ and point B with coordinates $(k, 24)$ where k is a negative constant. What is the value of the cosine of $\angle AOB$?